



PROJECT REPORT

On

DESIGN AND IMPLEMENTATION OF A MULTI-TARGET RADAR TRACKING AND VISUALIZATION ENGINE FOR ASTERIX CAT240 STREAMS

Submitted in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY In COMPUTER SCIENCE ENGINEERING

From

INDIRA GANDHI INSTITUTE OF TECHNOLOGY, SARANG

Submitted by

Mr. Kshirod Kumar Sahoo (Regd. No:- 2301105246)

Under the guidance of

Dependra S Parihar
Sc'E', LRDE

18th May 2026 to 10th July 2026

DECLARATION

I, **Kshirod Kumar Sahoo**, student of Bachelor Of Technology in Computer Science and Engineering at Indira Gandhi Institute Of Technology, Sarang hereby declare that the project entitled “**DESIGN AND IMPLEMENTATION OF AN MULTI-TARGET RADAR TRACKING AND VISUALIZATION ENGINE FOR ASTERIX CAT240 STREAMS**” has been carried out in the organization, DPSRR Division, LRDE, Bengaluru under the guidance of **Mr. DEPENDRA S PARIHAR (Sc ‘E’)**, DPSRR Division & **Dr. SRINIVAS SETHI**, Head Of The Department, Computer Science Engineering and Applications, Indira Gandhi Institute Of Technology, Sarang, in partial fulfillment of requirement for the award of degree, **BACHELOR OF TECHNOLOGY** in Computer Science and Engineering And Applications of Indira Gandhi Institute Of Technology, Sarang.

I also declare that, to the best of my knowledge & belief, the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this by any student.

Place: Bengaluru

Kshirod Kumar Sahoo

Date: 10/07/2026

ACKNOWLEDGEMENT

A student collects accolades in the form of grades for the succeed in his endeavors and this success depends on adequate preparation and indomitable determination and the most important of all the support he receives from his guide. So, the accolades we earn for this making.

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ABSTRACT

The project focuses on Radar system continuously generate large volume of video data, making it difficult to identify and track moving targets accurately in the presence of noise, clutter and measurement uncertainty. This project presents the design and implementation of a radar target Design and implementation of a multi target radar raw video data extraction and tracking system using ASTERIX CAT240 radar video data. To achieve reliable tracking, the detected measurements are passed through a Kalman Filter, which predicts the target's future position and continuously corrects its estimate whenever new radar measurement become available.

The Challenge: Radar target tracking involves several practical and technical challenges. One of the primary difficulties lies in processing raw radar video data, which contains thousands of amplitude values representing reflections from various objects within the radar's coverage area. Choosing a threshold that is too low may result in numerous false detections, while selecting a threshold that is too high may cause genuine targets to be missed. Also face challenges is the inherent uncertainty present in radar measurements. Every radar observation contains small errors caused by sensor limitations, signal propagation effects and external environment conditions. As a result the measured position of a target often fluctuates even when the target is moving smoothly. It also face challenges related to coordinate transformation and motion estimation.

Project Objective: The primary goal of this project is to develop a reliable radar target Design and implementation of a multi target radar raw video data extraction and tracking system capable of processing ASTERIX CAT240 radar video data and accurately estimating the position and movement of detected targets. The project aims to convert raw radar packets into meaningful tracking information by extracting radar parameters, identifying target echoes, estimating target location, and presenting the results through an intuitive graphical radar display. Another important objective is to demonstrate the practical application of the Kalman filter in radar tracking systems. The filter predicts the future position of a moving target based on its previous motion and continuously update its estimate not only the target position but also its target speed, heading and overall movement pattern.

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CHAPTER-1

INTRODUCTION

1.1 About the Organization:

The Defence Research and Development Organization (DRDO) is an agency of the Republic of India, responsible for the development of technology for use by the military, headquartered in New Delhi, India. It was formed in 1958 by the merger of the Technical Development Establishment and the Directorate of Technical Development and Production with the Defence Science Organization. DRDO has a network of 52 laboratories which are deeply engaged in developing defence technologies covering various fields, like aeronautics, armaments, electronic and computer sciences, human resource development, life sciences, materials, missiles, combat vehicles development and naval research and development. The organization includes more than 5,000 scientists and about 25,000 other scientific, technical and supporting personnel.

One of the establishments of DRDO is the Electronics and RADAR Development Establishment (LRDE) is a laboratory of the Defence Research & Development Organization (DRDO). Located in C.V. Raman Nagar, Bangalore its primary function is research and development of RADARs and related technologies. LRDE is sometimes mis-abbreviated as "ERDE". To distinguish between "Electrical" and "Electronic", the latter is abbreviated with the first letter of its Latin root (lektra).

1.1.1 Vision

To create a centre of excellence in design and development of radars and related Technologies.

1.1.2 Mission

Develop Radar systems and related technologies to cater to the needs of Services. Enhance the infrastructure, knowledge base and technologies for achieving self reliance.

LRDE has its genesis in the Inspectorate of Scientific Stores created in 1939 at Rawalpindi which was redesignated as Technical Development Establishment (Instruments and Electronics) in 1946 and located at Dehradun. In the year 1958 with bifurcation into inspection and R&D, the Electronics Research and Development Establishment was formed at Bangalore with men and material inherited from TDE (I&E).



1.1.3 Historical Background

The genealogy of LRDE originates from the Inspectorate of Scientific Stores created in 1939 at Rawalpindi. This was re-christened as TDE (Instruments & Electronics) in early 1946 and relocated in Dehradun. With electronic industrial complexes sprouting in the salubrious Bangalore outskirts, the Electronics part of TDE (I&E) branched off to settle in this seminal city in 1955. Coinciding with the promotion of DRDO, the then TDE (E) was bifurcated into Inspectorate of Electronics Equipments (ILE) and LRDE on 01 Jan 1958 to meet the

aspiration of Defence Services. LRDE moved to the present serenial location in 1986 to find place for expansion and growth.

Born in the barracks of High Grounds in Jan 1958 as Electronic Research and Development Establishment, got the present name Electronics & Radar Development Establishment in Jun 1962. With the nucleus personnel, equipment and material inherited from parental TDE (E), initial emphasis was laid to undertake such tasks as investigation of defects in equipment or necessary modifications required to improve the performance of equipment covering the entire spectrum of defence electronics.

LRDE scientists have been always ahead of their time. Phased Array design, Engineering of hardware and development of PCMs for Multimode Radar paved the way to develop weapon control radar Rajendra for Akash Weapon System. INDRA-I, INDRA-II with Pulse Compression and Rajendra radars rolled out in this era of uncertainties and challenges.

The unprecedented growth of LRDE during the second half of the Golden Jubilee period is due to multitude of enticing or compelling scenarios. Needless to say that the design, development and delivery of INDRA series radars, followed by Rajendra Phased Array Radar, WLR, BFSR-SR, 3D CAR, ROHINI, REVATHI, 3D-TCR, BHARANI, ASLESHA, XV-2004 and still counting, is the true reflector of maturity of LRDE, not only in the mastering of ground based, airborne and ship borne complex radar systems, but also practice of concurrent system engineering, R&D Industry Partnership and Professional approach to Technology & Project Management. The users unflinching faith has transformed LRDE as a force to reckon with. Many of such radars are now accepted or inducted by the users.

LRDE transformed itself as a matured, dependable system house of Radar, adopting modern Project management Practices, promoting industries as partners and collaborators, and developing human resources. Major thrust was given for self-reliance in technologies, components and industrial relationships.

With the activities directed towards design and development of complex, state-of-the-art electronically scanning TR module based active array ground based and airborne radars; it is a matter of time when the world will witness the marvelous achievements of LRDE in digital array radars.

1.1.4 Quality Policy

LRDE is committed to the indigenous development of modern radar systems through an effective Quality Management System.

1.2 Areas of Work

- **Design and Development of Radar Systems**

Army

- Multifunction Phased Array Radar and 3D Surveillance Radar for Akash Missile Weapon System
- Low Level Light weight 2D Radar for mountainous terrain Air Defence
- 3D -Tactical Control Radar for Air Defence
- Short Range Battle Field Surveillance Radar
- Weapon Locating Radar
- Multi Mission Radar (MMSR)
- FOPEN Radar
- Through wall detection Radar

- Ground Penetration Radar

Air Force

- Multifunction Phased Array Radar and 3D Surveillance Radar for Akash Missile Weapon System
- Active Phased Array Radar for AEW&C
- Low level 2D Air Defence Radar
- 3D Low Level Light Weight Radar
- 3D Medium Range Surveillance Radar for Air Defence
- 4D Active Array Medium Power radar for AD role
- Airborne Electronically Scanned Array Radar for Tejas Mark II
- Ground Controlled interception
- SAR for UAVs

Navy

- Maritime Patrol Radar for fixed and Rotary Wing Aircraft
- Maritime Patrol Radar with RS and ISAR
- 3D Medium Range Surveillance Radar for ASW Corvettes
- Multifunction Phased Array Radar for Air Defence Ship
- Maritime Patrol Airborne Radar for UAV
- Coastal Surveillance Radar (CSR)

Development of Radar Technologies

- Slotted Waveguide Antenna in S, X and Ku Band
- Patch Array Antenna in S, C, X and Ku Band
- Multibeam Antenna in S Band
- T/R Modules in L, S, C and X Band for Active Aperture Arrays
- Semi Active Aperture Arrays

- MMICs for T/R Modules
- Central Unit
- Space Time Management
- Programmable Signal Processors
- FPGA based Signal Processors
- Radar Data Processors
- High Average Power TWT based Transmitters
- Compact Light Weight TWT Transmitters.
- High purity sources
- Multichannel Digital receivers
- Digital Beam Formers
- Solid State Transmitters
- Radar Imaging through ISAR
- Automatic Target Classification
- Through Wall Imaging Radar (TWIR)
- Ground Penetration Radar (GPR)
- Array Group Receiver

1.4 Preface:

I being a student of Computer Science Engineering department choose this topic, **Design and Implementation of a Multi-Target Radar Tracking and Visualization Engine For ASTERIX Cat-240 Streams**, which is relevant and confined to my field. I am conducting this project at a well-recognized organization which is working for the Government of India. Under the guidance of my guide I am working on developing a radar target Design and implementation of a multi target radar raw video data extraction and tracking system using ASTERIX CAT240 radar video data. The project aims to:

- Develop an efficient radar target Design and implementation of a multi target radar raw video data extraction and tracking system capable of processing ASTERIX CAT240 radar video data
- Improve tracking accuracy by minimizing the effects of measurement noise and uncertainty
- Radar measurements from polar coordinates into Cartesian coordinates to facilitate motion estimation and graphical visualization

Radar technology, short for Radio Detection and Ranging is a corner stone of modern scientific and military advancements. It operates by emitting radio waves that travel through space, interact with objects in their path, and return as echoes. These echoes hold critical information about the presence, location and movement of objects, making radar indispensable for various applications.

This project addresses the vital need for efficient radar target Design and implementation of a multi target radar raw video data extraction and tracking system using ASTERIX CAT240 radar video data. After detecting the target, the measured range and azimuth are converted from polar coordinates to the Cartesian coordinates(X and Y).These measurements are then fed into a discrete Kalman Filter one of the most widely established recursive estimation algorithms to predict dynamic target trajectories and estimate velocities. Through a meticulous analysis of radar data available at the D-PSRR division of DRDO's Electronics and Radar Development Establishment (LRDE), this work demonstrates a robust software pipeline capable of transforming raw digitized telemetry into deterministic, real time target tracks.

The significance of this endeavor extends beyond its technical intricacies; it holds the potential to significantly booster national security.

CHAPTER-2

LITERATURE REVEIW

2.1 Literature Review

Design and implementation of a multi target radar raw video data extraction and tracking have been active areas of research for several decades due to their importance in applications such as air traffic control, defence surveillance, maritime navigation, autonomous vehicles, and weather monitoring. The primary objective of a radar tracking system is to accurately detect moving targets, estimate their positions, and continuously monitor their trajectories despite the presence of noise, clutter and measurement uncertainties. Over the years, researchers have developed numerous signal processing and estimation techniques to improve the reliability and accuracy of radar tracking system. Once the target has been detected, accurately estimating its position becomes another important challenge. Radar measurement naturally provides information in polar coordinates, consisting of range and azimuth, while most tracking algorithms operate more efficiently in Cartesian coordinates. Tracking a moving target requires estimating its current state while predicting its future motion. Among the numerous tracking algorithms available, the Kalman Filter has emerged as one of the most widely used and successful estimation techniques for linear dynamic systems. Introduced by Rudolf E. Kalman in 1960, the Kalman Filter provides an efficient recursive solution for estimating the state of a system from noisy observation. Several researchers have extended the classical Kalman Filter to handle more complex tracking scenerios. The extended Kalman Filter was developed to estimate non linear systems by linearizing the non linear equations around the current operating point.

The ASTERIX CAT240 standard has become increasingly important for transmitting radar video data between surveillance sensors and processing systems. Unlike traditional radar reports that only provide processed target information, CAT240 allows the transmission of complete radar video, enabling developers to implement custom detection, filtering, and tracking algorithms.

2.1 Existing system:

Existing Radar Systems:

- Surveillance Radars: These radars provide wide-area coverage for detecting vessels.
- Navigation Radars: Primarily used for ensuring safe navigation for ships and aircrafts.
- Military Radars: Critical for defence and target tracking applications.
- Weather Radars: Essential for meteorological applications, including weather forecasting.
- Phased Array Radars: Known for rapid beam scanning and flexibility in radar beam control.

In modern aviation and defence infrastructures, high fidelity radar tracking systems are well established. The underlying radar hardware is already fully capable of transforming electromagnetic waves, receiving reflections, and performing initial hardware level digitization. These systems standardize their raw, digitized video data into protocols like the ASTERIX CAT240 format, which contains pre detected target reflections mixed with environmental noise and metadata.

In a standard operational workflow, this raw video data is typically fed into dedicated hardware processors or legacy display units that parse the streams using standard, rigid threshold boundaries. While these traditional pipelines are highly efficient at handling consistent streams, they often pass the burden of track smoothing, noise filtering, and data association down the line, or they rely on hardcoded threshold that struggle when background noise or clutter levels change dynamically.

2.1.1 Drawbacks:

- Direct reliance on Unrefined Measurements: Legacy systems often take raw, unfiltered positional data directly from the video packet and map it straight to an operator's screen. Because these raw observations contain natural sensor uncertainty, target positions fluctuate aggressively between consecutive radar scans. The jitter produces erratic, unstable trajectories and makes it difficult to calculate accurate real time range azimuth and speed.
- Closed Architectures and Hardware Lock-in: Instead of providing access to the raw ASTERIX CAT240 video packets, they only output finalized, black-box target lists. This deprives developers and engineers of the flexibility needed to test custom feature-extraction techniques, custom filtering algorithm.
- Track Discontinuity During Signal Fades: Standard video processing setups less predictive, recursive state estimation. Without a tracking framework to estimate a target's future state or bridge temporary data gaps, target tracks instantly fracture. The moment a target experiences a localized signal fade or passes through a noisy patch, its tracking history is lost and split into a completely challenging entity.

2.2 Objective of the study:

This project addresses the need for an efficient design and implementation of a multi target radar raw video data extraction and tracking system capable of processing ASTERIX CAT240 radar video data and accurately monitoring the movement of detected targets. The primary objective is to extract meaningful information from raw radar packets by identifying valid target echoes,

estimating their positions, and continuously tracking their motion across successive radar sweeps. Since radar measurements are often affected by noise, clutter, and environmental uncertainties, the study focuses on improving tracking accuracy through the implementation of a Kalman Filter. The Kalman Filter predicts the future state of a target based on its previous motion and refines these predictions whenever new radar measurements become available, resulting in smoother and more reliable target trajectories. Another important objective of the study is to implement an effective target association mechanism that correctly matches newly detected radar measurements with existing target tracks. The system assigns a unique Track ID to each detected object and maintains its identity throughout multiple radar rotations using a nearest neighbour data association technique. It also incorporates track management features such as track creation for newly detected targets, prediction during temporary target loss (track coasting), and automatic deletion of inactive tracks after three consecutive missed radar sweeps. These features ensure continuous and robust target tracking while minimizing false tracks and duplicate target identities. The study also aims to convert radar measurements from polar coordinates (range and azimuth) into Cartesian coordinates (X and Y), enabling efficient motion estimation, velocity calculation, and graphical visualization. Furthermore, detected targets are classified into different intensity levels based on their radar echo strength, allowing operators to distinguish between weak, medium, and strong targets. To enhance situational awareness, the project provides a user-friendly graphical interface developed using OpenCV, where the radar sweep, detected targets, Kalman Filter estimated positions, movement vectors, unique Track IDs, and target intensity levels are displayed in real time. The research has been carried out using radar data obtained from the D-PSRR Division at the Defence Research and Development Organisation (DRDO), specifically the Electronics and Radar Development Establishment (LRDE). By

combining radar signal processing, target detection, Kalman Filter–based state estimation, and intelligent track management, the proposed system demonstrates a practical and reliable solution for modern radar surveillance applications. The significance of this work extends beyond technical implementation, as it contributes to enhancing radar situational awareness, improving surveillance accuracy, and strengthening capabilities in defence, border security, airspace monitoring, and other critical national security applications.

2.3 Proposed system:

The proposed system is an efficient design and implementation of a multi target radar raw video data extraction and tracking solution developed to process **ASTERIX CAT240 radar video data** and provide accurate real-time target surveillance. The packet structure is like :

CAT = 240	LEN	FSPEC	Items of the video record
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It is implemented using **Python**, **OpenCV**, and a **Kalman Filter** to detect, identify, and continuously track moving targets across successive radar sweeps. Unlike conventional radar systems that display only raw echoes, the proposed system assigns a unique **Track ID** to each detected target and maintains its identity throughout the tracking process.

The system begins by reading CAT240 radar packets and extracting important parameters such as azimuth, cell duration, and radar video amplitudes. Using the cell duration, it calculates the radar range resolution, while threshold is applied to separate genuine target echoes from background noise. Neighbouring radar cells with significant amplitudes are grouped into clusters, and the weighted centroid of each cluster is calculated to determine the target's position. Based on the peak amplitude, targets are classified into **Low**, **Medium**, or **High** intensity categories.

The detected target coordinates are converted from **polar coordinates (range and azimuth)** to **Cartesian coordinates (X and Y)**, which are then used by the Kalman Filter for tracking. The Azimuth and Range is calculated from the packets by the following formulas :

DPSRR

START_AZ Start azimuth of the cells group
LSB = $360/2^{16}$, Range: [0;360]

END_AZ End azimuth of the cells group
LSB = $360/2^{16}$, Range: [0;360]

$$D = \text{CELL_DUR}_{(\text{in seconds})} \times (\text{START_RG} + \text{NU_CELL} - 1) \times c / (2.)$$

where $c = 299\,792\,458$ m/s: light celerity.

The filter predicts the target's future position and updates its estimates whenever new measurements are received. A **Nearest Neighbour Data Association** algorithm matches new detections with existing tracks. If no suitable match is found, a new track with a unique Track ID is created.

To improve tracking reliability, the system also incorporates **track management**. Targets that are temporarily not detected are maintained through **track coasting**, where their positions are predicted until new measurements become available. If a target remains undetected for **three consecutive radar sweeps**, its track is automatically deleted to remove outdated or false tracks.

Finally, the processed information is displayed on an OpenCV based radar interface, showing the radar sweep, detected targets, Track IDs, estimated positions, movement vectors, and intensity levels. By combining threshold, Kalman Filter prediction, nearest neighbour association, and efficient track management, the proposed system provides an accurate and reliable solution for modern radar surveillance applications such as air surveillance, maritime monitoring, border security, and defence systems.

2.3.1 Advantages:

- **Accurate Target Tracking:** The Kalman Filter continuously predicts and updates target positions, resulting in smooth and reliable tracking even when radar measurements contain noise.
- **Unique Target Identification:** Every detected target is assigned a unique Track ID, allowing the system to distinguish and monitor multiple targets simultaneously without confusion.
- **Reduced False Detections:** threshold effectively separates genuine target echoes from background noise and clutter, improving the overall detection accuracy.

- **Continuous Tracking:** The system maintains target tracks across successive radar sweeps, ensuring that moving objects are tracked consistently over time.
- **Handles Temporary Target Loss:** Through track coasting, the system continues predicting a target's position even if it is temporarily missed, reducing unnecessary track loss.
- **Automatic Track Management:** Tracks that remain undetected for three consecutive radar rotations are automatically removed, keeping the radar display clean and free from outdated information.
- **Efficient Data Association:** The Nearest Neighbour algorithm correctly matches new radar measurements with existing tracks, minimizing duplicate or incorrect target assignments.
- **Real-Time Visualization:** The OpenCV-based interface provides a clear display of radar sweeps, detected targets, Track IDs, movement vectors, and target intensity levels, making monitoring easier for the operator.
- **Scalable and Flexible:** The system can efficiently handle multiple targets and can be further enhanced with advanced tracking or data association algorithms in the future.
- **Wide Range of Applications:** The proposed system can be used in air surveillance, maritime monitoring, border security, defence systems, and other radar-based monitoring applications where accurate target tracking is essential.

2.4 Requirements Specifications:

The successful implementation of the proposed radar target “Design and implementation of a multi target radar raw video data extraction and tracking system” requires software resources. These requirements ensure that the system can efficiently process ASTERIX CAT240 radar data, perform multi target detection and tra, and display the results in real time.

2.4.1 Software requirements:

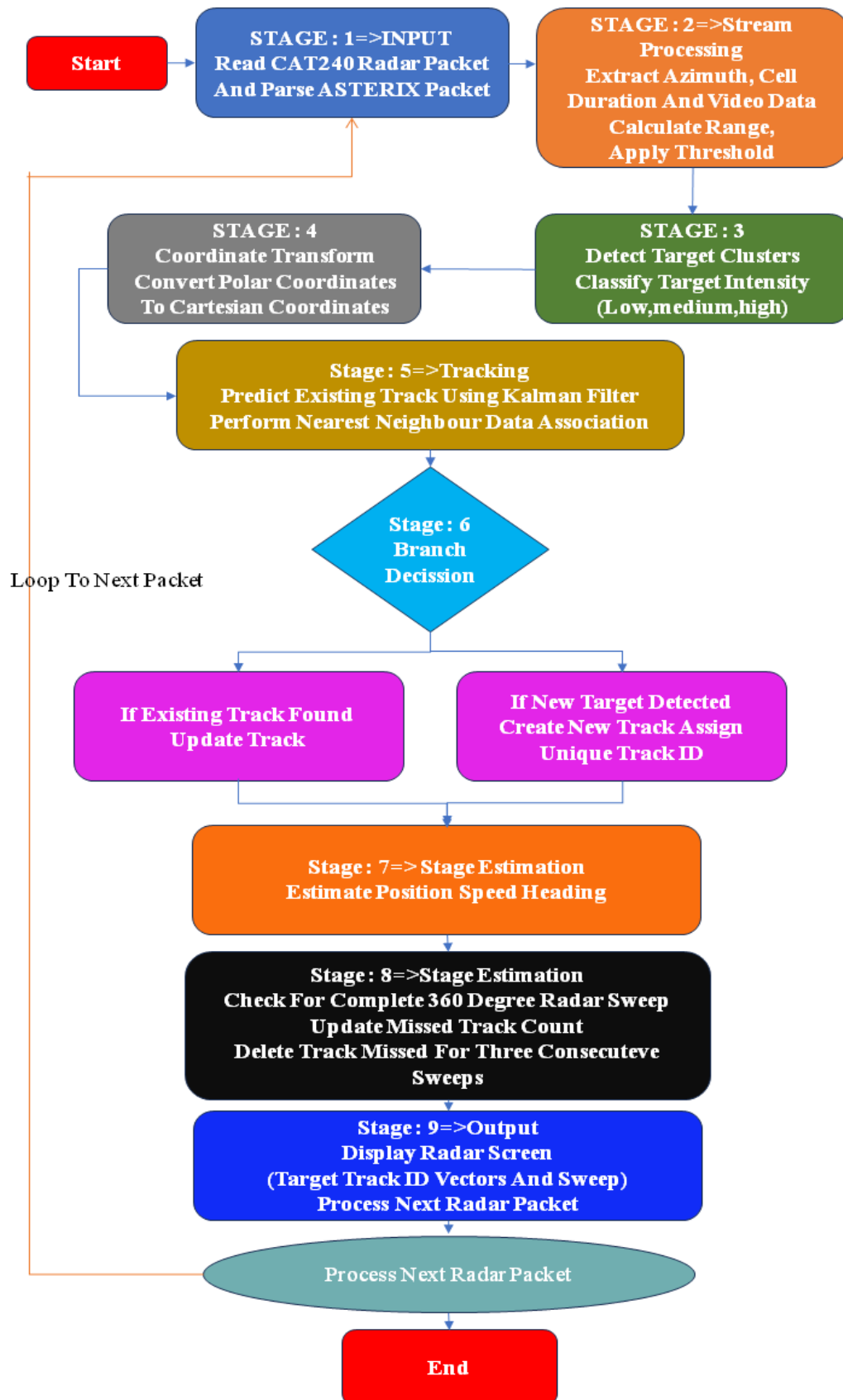
The proposed system is implemented entirely in python due to its simplicity, extensive scientific computing libraries and excellent support for image processing and numerical analysis.

- **Operating System:** Windows 7
- **Programming Language:** Python 3.7.3
- **Development Environment:** Jupyter Notebook, Notepad, Anaconda Prompt
- **Libraries used:**
 - *OpenCV* : Used for creating radar display, drawing targets, Visualizing tracks, handling graphical output.
 - *NumPy* : Used for numerical computations, matrix operations and Kalman Filter Calculations.
 - *Math* : Used for trigonometric functions, coordinate conversion, speed calculation, heading estimation.
 - *Regular Expression* : Used to extract numerical values from ASTERIX CAT240 packet files.
 - *OS Module* : Used to access sequential radar packet files and manage file operations.

CHAPTER-3

MODEL DESIGN

3.1 Flow diagram:



3.2 Description

- The process begins by reading the ASTERIX CAT240 radar packet files. These packets contain raw radar video data along with the necessary information required for Design and implementation of a multi target radar raw video data extraction and tracking.
- The radar packet is parsed to extract important parameters such as the start azimuth, end azimuth, cell duration, and radar video amplitude values. These parameters form the basis for further processing.
- The extracted azimuth values determine the radar scan direction, while the cell duration is used to calculate the range resolution. The radar video data contains the signal amplitudes that indicate possible target echoes.
- Using the cell duration, the system calculates the distance represented by each radar cell. This enables the accurate estimation of the target's distance from the radar.
- An threshold is applied to distinguish genuine target echoes from background noise and clutter. This improves the reliability of target detection under different operating conditions.
- Radar cells with amplitudes above the threshold are grouped into clusters. Each cluster represents a potential target detected by the radar.
- The weighted centroid of each cluster is calculated to estimate the target's exact position more accurately than using individual radar cells.
- Based on the peak signal amplitude, the detected target is classified as **Low**, **Medium**, or **High** intensity. This helps identify the relative strength of the radar echo.
- The detected target's position is converted from polar coordinates (range and azimuth) into Cartesian coordinates (X and Y), making it suitable for target tracking and visualization.

- The Kalman Filter predicts the current position of every active target based on its previous position and velocity. This prediction helps maintain smooth and continuous tracking.
- The predicted tracks are compared with the newly detected target. If the detection matches an existing track, it is associated with that track; otherwise, it is treated as a new target.
- If a matching track is found, the Kalman Filter updates the target's state using the new measurement. If no match exists, the system creates a new track and assigns it a unique Track ID.
- After the track is updated, the system calculates the target's estimated position, speed, and direction of movement, providing a more accurate representation of its motion.
- At the end of every complete radar rotation, the system checks whether each active target has been detected during the current sweep.
- Targets that are not detected during a sweep are marked as missed, and their missed count is increased while the Kalman Filter continues predicting their positions.
- If a target remains undetected for three consecutive radar sweeps, its track is automatically removed to prevent outdated or false tracks from remaining in the system.
- The processed results are displayed on the radar interface, including detected targets, Track IDs, estimated positions, movement vectors, intensity levels, and the rotating radar sweep.
- The system repeats the same sequence for the next radar packet, allowing continuous real time Design and implementation of a multi target radar raw video data extraction and tracking until all radar data has been processed.

CHAPTER-4

IMPLEMENTATION

4.1 Python Algorithm Implementation

- i. Initialize the System:-creates Kalman Filter tracks. Initialize track pool, radar display and the parameters.
- ii. Read radar data:-Read sequential ASTERIX CAT240 packet files. Extract packet fields such as Azimuth, Cell duration etc.
- iii. Detect Targets:-Apply threshold to remove noise and compute amplitude weighted centroid for the clusters.
- iv. Convert Cartesian coordinates:-Convert detected target range and azimuth into X-Y coordinates.
- v. Manage radar sweep:- Detect the completion of a radar sweep. Removes stale tracks and predict targets using Kalman Filter.
- vi. Track targets:- Associate detections with existing tracks using nearest neighbor matching. Create a new track if no existing track matches.
- vii. Display results:-Draw radar sweep, detected track, track IDs. Classify targets as HIGH, MEDIUM, LOW intensity using color coding.
- viii. Repeat:-Process next radar packet until all packets are processed .

4.2 Simulation Process

The Simulator was implemented to provide continuous sequence of data to Multi-Target Radar Tracking and Visualization Engine in a loop that mimics a live radar environment. Simulator sequentially provided data packets in a continuous manner updating a virtual 360 degree on PPI scope Interface. A sweep boundary result mechanism tracked the radar's rotating azimuth. Whenever a full 360 degree rotation was completed, the system refreshed the visual history smoothly based on rotation speed. Once the objective was met with simulated data, real data was utilized for the verification & testing of algorithms.

4.3 Real Data Integration

The software architecture was made to handle standard ASTERIX CAT240 video formats in real-time. By converting raw binary packet structures into physical units i.e. digital radar video polar coordinates onto real world coordinates. Real radar data was collected through UDP and converted into packets and then the useful data has been collected .This approach ensured that the clustering and filtering algorithms function identically whether work against simulated data or live data.

CHAPTER-5

RESULTS

4.1 Software simulation results:

The software successfully processes simulated data by parsing simulated ASTERIX CAT240 packets. An threshold applied, while a spatial clustering module combines raw returns into single target centroids. These calculated coordinates are produced by Kalman Filter tracker, which automatically assigns persistent target IDs and generate range, azimuth & speed. Additionally, the tracking logic successfully maintains target continuity by coasting tracks through temporary signal graph.

4.1.1 Real Data results:

The proposed radar target design and implementation of a multi target radar raw video data extraction and tracking system was successfully implemented and tested using sequential **ASTERIX CAT240 radar packet** data. During the simulation, the system accurately parsed the radar packets, extracted the required radar parameters, and detected valid targets by applying threshold. Background noise and unwanted echoes were effectively filtered, resulting in reliable target detection.

The **Kalman Filter** successfully predicted and updated the position of moving targets across successive radar sweeps. Even when temporary target loss occurred, the tracker continued estimating the target's position through track coasting, ensuring smooth and continuous tracking. The **Nearest Neighbour Data Association** algorithm correctly associated new radar measurements with existing tracks, while newly detected objects were assigned unique **Track IDs**.

The simulation also verified the track management mechanism. Targets that were continuously detected maintained the same Track ID throughout the simulation, whereas inactive targets that remained undetected for three consecutive radar sweeps were automatically removed from the tracking database. This prevented outdated or false tracks from appearing on the radar display.

The graphical interface developed using **OpenCV** successfully displayed the radar sweep, detected targets, Kalman Filter estimated positions, Track IDs, movement vectors, and target intensity levels in real time. The system demonstrated stable performance, accurate target localization, smooth trajectory estimation, and efficient track management throughout the simulation. Overall,

the obtained results confirm that the proposed system provides an effective and reliable solution for real-time radar target Design and implementation of a multi target radar raw video data extraction and tracking.

00000000	f0	04	23	eb	a8	36	22	02	00	08	a4	e1	71	18	71	59
00000010	00	00	00	00	00	00	00	c0	00	04	04	20	00	04	20	10
00000020	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000030	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000040	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000050	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000060	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000070	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000080	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000090	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000000a0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000000b0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000000c0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000000d0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
000000e0	00	00	00	00	00	00	00	00	10	00	00	00	00	00	00	00
000000f0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000100	00	00	00	10	00	00	00	00	00	00	00	00	00	00	00	00
00000110	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000120	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000130	10	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000140	00	00	00	00	00	00	00	00	00	00	00	00	00	00	10	00
00000150	00	00	00	00	00	00	00	00	00	00	00	00	00	00	10	00
00000160	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000170	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
00000180	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

Figure 1: ASTERIX CAT240 Raw Video Data

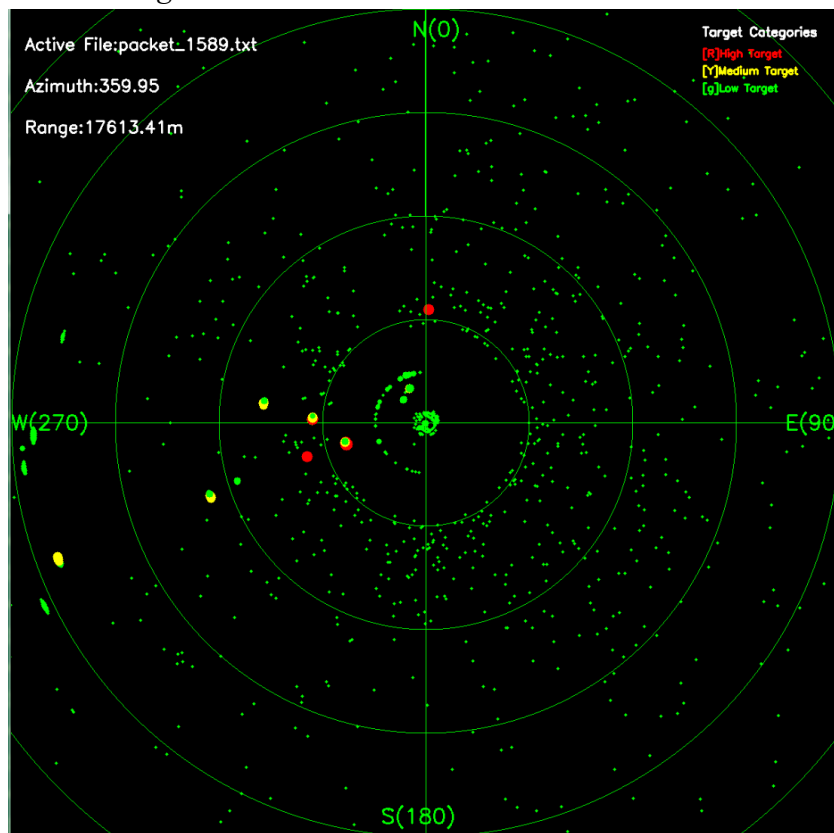


Figure 2: visualization of raw video data

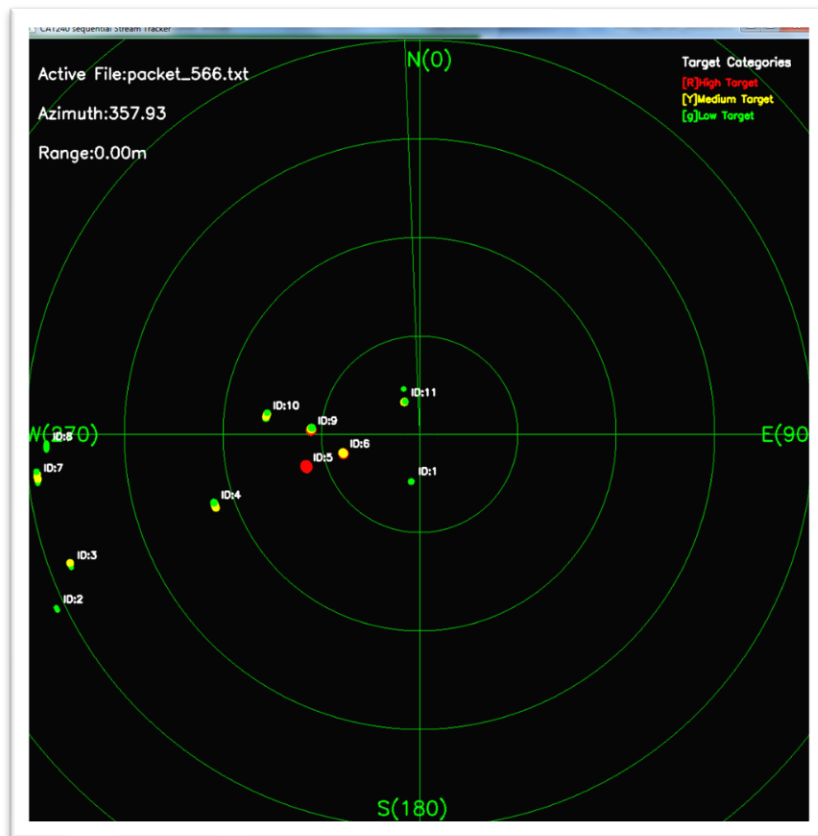


Figure 3: Visualization after clearing low amplitude data

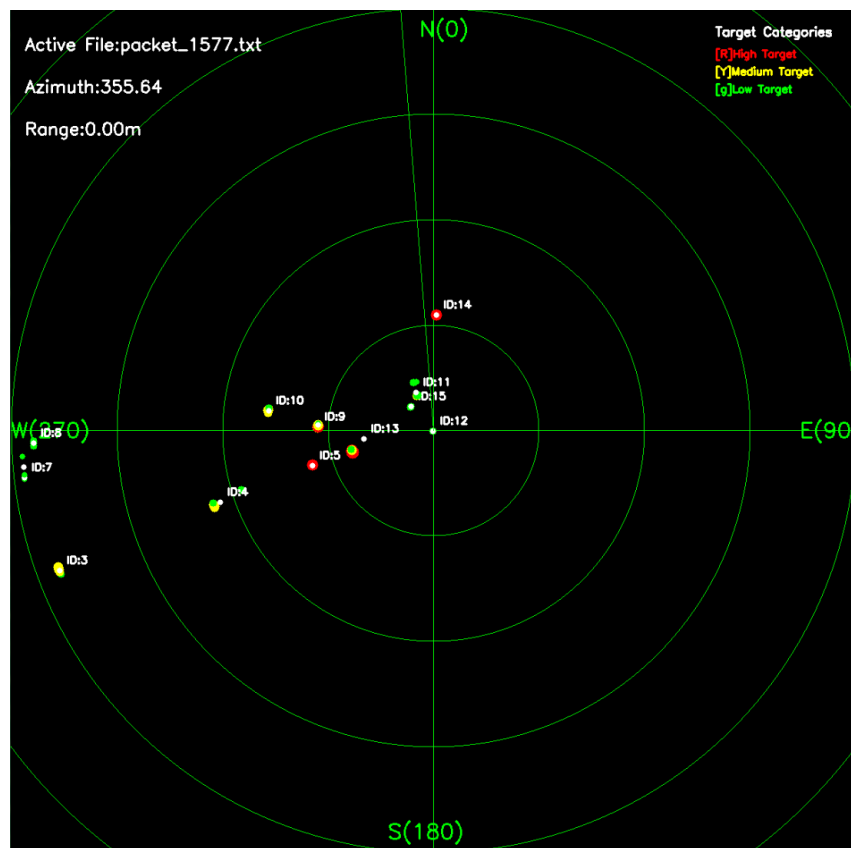


Figure 4: Kalman Filter Prediction (Represented by white blobs)

```

[Target Locked] ID:9 : Intensity:HIGH

Actual    =>Range:  6555.3m : Azimuth:272.66deg : X: -6548.2 : Y:  304.8
Estimated =>Range:  6554.4m : Azimuth:272.47deg : X: -6548.3 : Y:  282.6

[Target Locked] ID:10 : Intensity:LOW

Actual    =>Range:  9426.5m : Azimuth:276.72deg : X: -9361.8 : Y:  1102.8
Estimated =>Range:  9424.0m : Azimuth:276.73deg : X: -9359.0 : Y:  1104.4

[Target Locked] ID:15 : Intensity:LOW

Actual    =>Range:  1841.9m : Azimuth:316.42deg : X: -1269.6 : Y:  1334.4
Estimated =>Range:  1841.9m : Azimuth:316.42deg : X: -1269.6 : Y:  1334.4

[Target Locked] ID:11 : Intensity:LOW

Actual    =>Range:  2377.5m : Azimuth:335.89deg : X: -971.1 : Y:  2170.1
Estimated =>Range:  2366.7m : Azimuth:335.85deg : X: -968.3 : Y:  2159.6

[Target Locked] ID:12 : Intensity:LOW

Actual    =>Range:    57.6m : Azimuth:206.42deg : X:   -25.6 : Y:   -51.5
Estimated =>Range:    57.6m : Azimuth:206.62deg : X:   -25.8 : Y:   -51.5

[Target Locked] ID:2 : Intensity:LOW

Actual    =>Range: 24491.3m : Azimuth:244.19deg : X:-22048.4 : Y:-10662.7
Estimated =>Range: 24491.3m : Azimuth:244.19deg : X:-22048.4 : Y:-10662.7

[Target Locked] ID:3 : Intensity:LOW

Actual    =>Range: 22678.1m : Azimuth:249.31deg : X:-21215.7 : Y: -8012.0
Estimated =>Range: 22678.1m : Azimuth:249.32deg : X:-21217.4 : Y: -8007.4

```

Figure 5: Extracted and Estimated Range, Azimuth, X, Y

Figure 1 displays the ASTERIX CAT240 Raw Video Data results. Figure 2 visualizes all the noises, clutters including the targets present in the raw video. Figure 3 visualizes the useful data means having the high amplitude data. Figure 4 visualizing the white blobs using Kalman Filter algorithm. Figure 5 represents the extracted and predicted Range, Azimuth, X, Y. Each data point is represented as a colored coordinate, with similar colors indicating data points.

CHAPTER-6

CONCLUSION

During my internship at **DRDO's Electronics and Radar Development Establishment (LRDE)**, I successfully designed and implemented of a multi target radar raw video data extraction and tracking system using **ASTERIX CAT240 radar video data**. The project focused on accurately tracking radar targets, reducing the impact of measurement noise, and maintaining continuous target tracking using the **Kalman Filter**. By incorporating threshold, target clustering, nearest-neighbour data association, and intelligent track management, the developed system was able to identify, track, and monitor moving targets reliably across successive radar sweeps.

The simulation results demonstrated that the proposed system effectively assigns unique Track IDs, predicts target positions during temporary detection loss, and automatically removes inactive tracks after consecutive missed radar rotations. The OpenCV based visualization further enhanced situational awareness by displaying the radar sweep, target positions, movement vectors, and intensity classifications in real time.

The knowledge and experience gained during this internship strengthened my understanding of **radar target detection, target tracking algorithms, Kalman filtering, and real-time visualization techniques**. Working with real radar data from the **Dte. of PSRR, LRDE** provided valuable exposure to practical defence technologies and their real-world applications.

In conclusion, the developed system provides a reliable and efficient solution for modern radar surveillance by improving target detection accuracy, maintaining continuous tracking, and supporting effective decision-making. The project has significant potential for applications in **air surveillance, maritime monitoring, border security, and defence systems**, where accurate and dependable target tracking is essential. It also provides a strong foundation for future enhancements, such as incorporating multiple target tracking algorithms, advanced data association techniques, and AI and ML based target classification to further improve the performance of radar surveillance systems.

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Thank You.